

Design review #2/3 – Level shifters and logic



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1 General information

This project will develop an instrumentation amplifier (INA) with programmable gain (through I2C / SPI interfaces) which achieves high accuracy (< 1% gain error), low noise (< 50nV/rtHz at 1kHz), and high common mode rejection (> 80dB). Design reviews #2 and #3 for this are described in [1] and [2] respectively. In parallel with that, a 6b tunable ring oscillator (RO) has been developed. Design reviews #2 and #3 for this are described in [3] and [4] respectively. As the level shifters and their accompanying logic are applicable to both the INA and RO, they are treated as separated IP which exist outside of the INA / RO hierarchies. As a result, they mandate separate design reviews.

Due to relatively small content required for the DR's of these blocks, both DR2 and DR3 and lumped together in this single doc.

1.1 Project goal

To show the capability of open source tools to develop industry quality ASICs, verified across industry standard conditions and taped out on silicon from a European fab (IHP).

To validate this goal, targeted performance will match that of an INA from a leading industry vendor (AD8223A [5]).

1.2 Specifications

Because this design review deals with level shifters and associated logic, specifications are purely functional across across full process (SS/FF/SF/FS), voltage ($\pm 5\%$) and temperature ($-40 \rightarrow 125\text{C}$) (i.e. PVT) conditions.

1.3 Purpose of Design review #2 (DR2) and Design review #3 (DR3)

Purpose of DR2 is to review final schematics before delivery to layout. Therefore, upon approval of DR2, schematics are "frozen" and layout commences.

Purpose of DR3 is to review final layouts before delivery to top level (i.e. chip level) integration. Therefore, upon approval of DR3, layouts are "frozen" and top level integration commences. Specifically, this DR3 refers to the layout freeze of the INA top level (layout freezes for blocks outside of the INA top level are detailed in separate DR3's).

1.4 Repository details

All material relating to this project can be found in the below publicly available repository:

https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main

As this document pertains to both DR2 and DR3, all material relating to it is found in the sub directory "Design_Review_3" (folder = LVL_SHFT).

As block schematics relate to all DR's, these are contained in the separate folder "Schematics_Layout".

Inside each folder here, lives the following views:

- Schematic (to be frozen at DR2)
- Symbol (to be frozen at DR2)
- Layout (to be frozen at DR3)

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Checked in schematic / symbol / layout for the level shifter core is shown in Figure 1 while that for the core and its accompanying logic is shown in Figure 2.

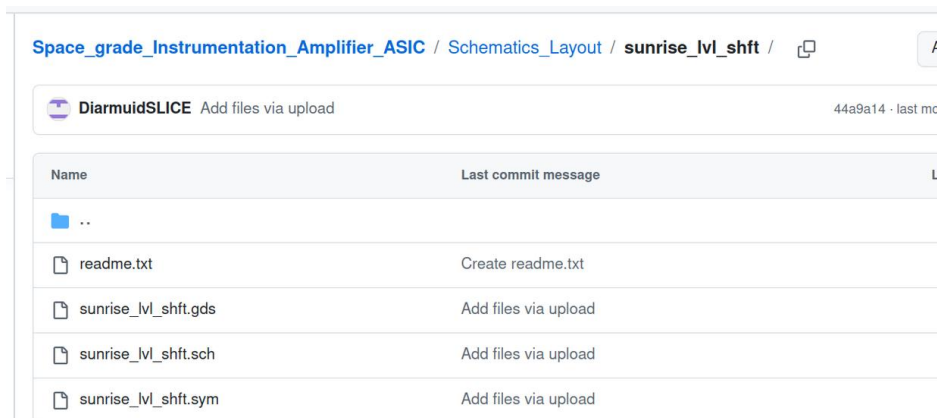


Figure 1 Checked in schematics for level shifter core

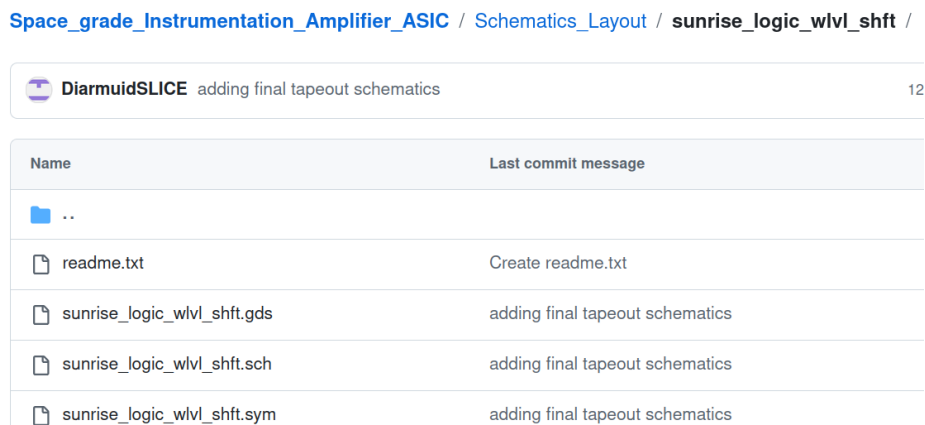


Figure 2 Checked in schematics for level shifter core with accompanying logic

1.5 Open source tools

The open source tools used for DR2 are summarized in Table 1, with links to install each tool provided as references.

Tool	Description
ngspice	Spice simulator [6]
xschem	Schematic / waveform viewer [7]
openVAF	verilogA compiler [8]
Octave	Post processing [9]
klayout	Layout viewer / editor [10]
magic	Pcell generator / PEX generation [11]
netgen	LVS [12]

Table 1 Open source tools used for DR2/3

1.6 Open source PDK

The open source process development kit (PDK) used for the design is from IHP, process option – SG13CMOS5L [13]. This is a derivative of their more general option – SG13G2 [14] as it excludes the following:

- Top metal layer 2
- Linear caps (MiM or MoM)
- Deep Nwell (DNW)
- HBT devices

The first 3 items impacted the design but not the 4th since the design is not RF. SG13CMOS5L option was chosen primarily because its tapeout date of 13/07/26 suited the project timeline. In addition, it is lower cost (max = €1.5k/mm² as opposed to €2.8k/mm² for SG13G2) and runs through the fab quicker (5 months instead of 6) to achieve a fab out date in 16/12/26 [15].

2 Top-Level design description

2.1 Architecture

Core level shifter IP was generously donated to this project from Roel Jordans [16]. Only one modification was made to this IP and that was replacing the customized input inverter with one from the std lib, for layout reasons. Resulting symbol and schematic are shown in Figure 3 and Figure 4.

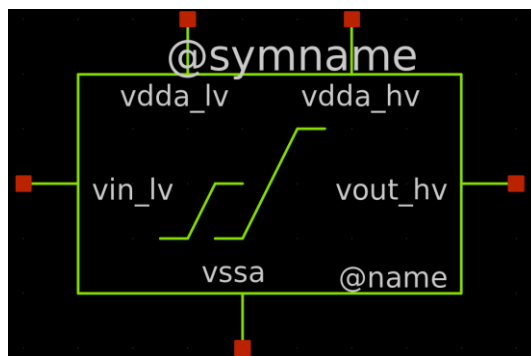


Figure 3 Core level shifter (LS) symbol

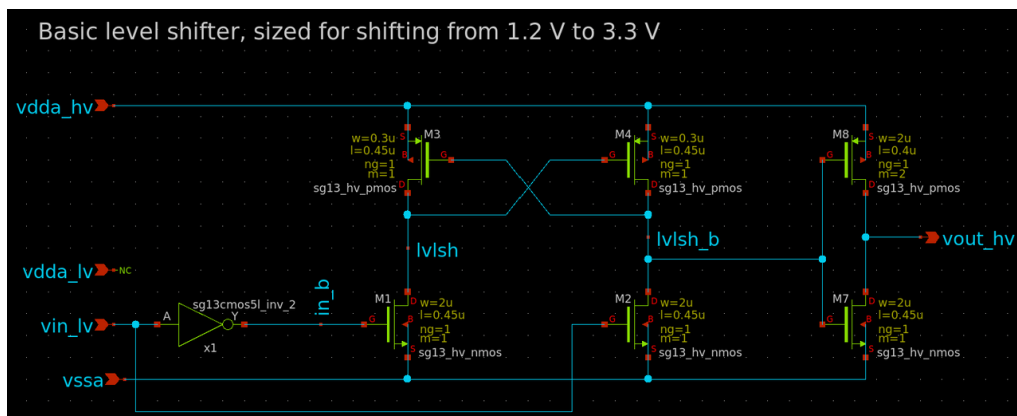


Figure 4 Core level shifter (LS) schematic

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As can be seen from Figure 4, the LS is a basic architecture with no reset. Having no reset necessitates correct power up sequencing to ensure vdda_lv comes up before vdda_hv. With no reset, if vdda_hv comes up before vdda_lv the LS can enter an undefined state leading to unwanted paths. As shown in [17], great care has been taken to ensure the correct power up sequence because of this.

At the top level, this level shifter is placed in a bank of 20 level shifters whose symbol and schematic are shown in Figure 5.

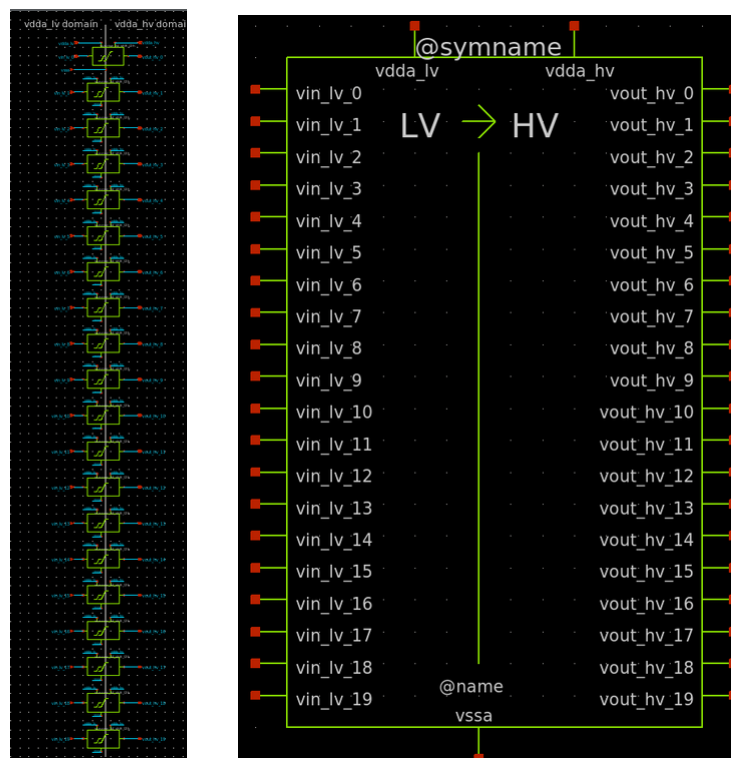


Figure 5 Level shifter bank symbol and schematic

As the function of these level shifters is to shift internal logic referenced to vdda_lv to vdda_hv, they are accompanied by an internal logic block, referred to vdda_lv, as shown in Figure 6.

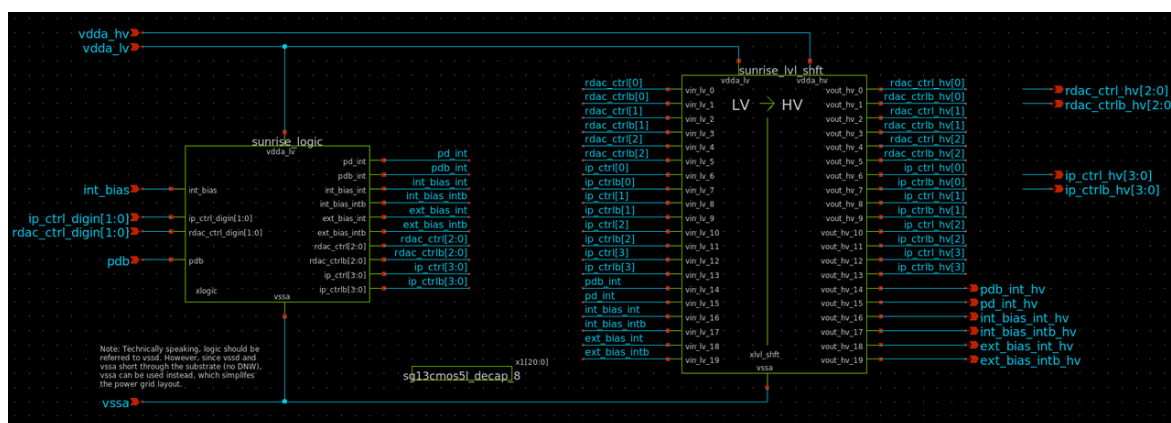


Figure 6 Level shifter bank and accompanying logic block

The accompanying logic comprises of the following sub-blocks_

- Block to handle powerup / down of the ASIC (Figure 7)
- Binary to one hot code block to select inputs to the ASIC (Figure 8)
- Decode block to set the INA gain (Figure 9)

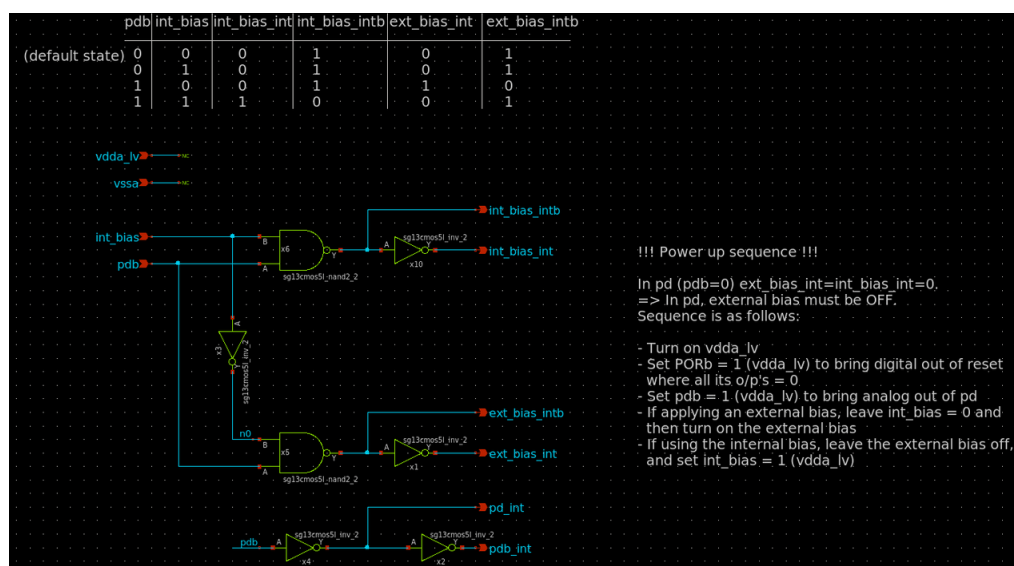


Figure 7 Powerup / down control

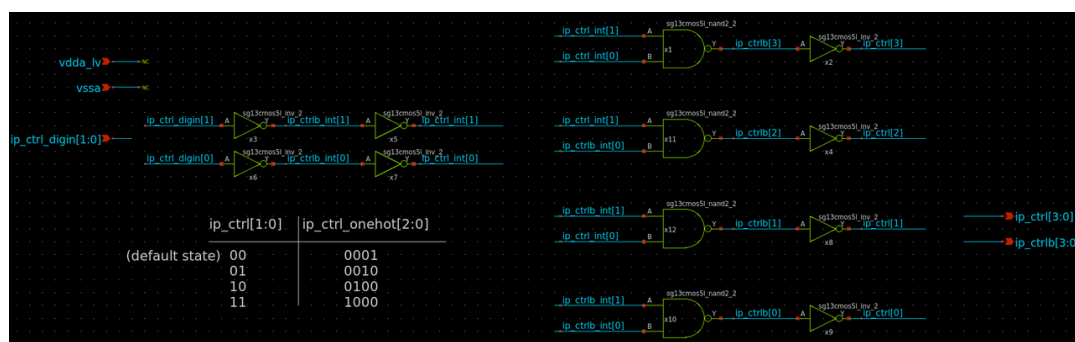


Figure 8 Binary to one hot input control

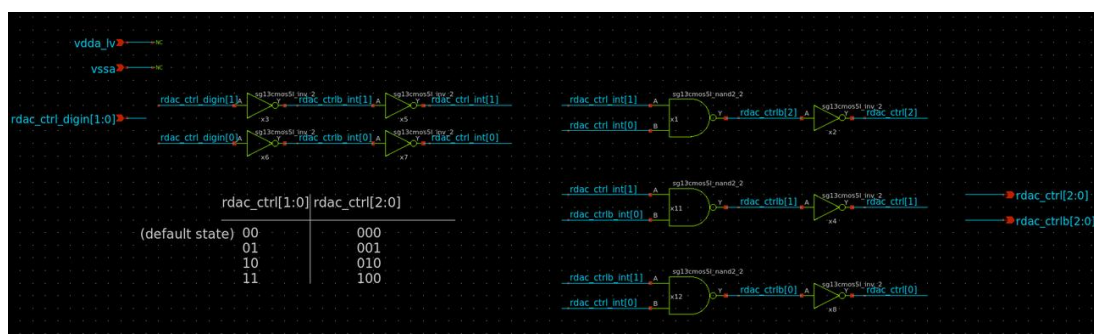


Figure 9 Decode block to set INA gain

3 Testbench description

3.1 Testbench details

Testbench used to verify the functionality of the level shifter core is shown in Figure 10 and checked into the below:

https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_3/LVL_SHFT

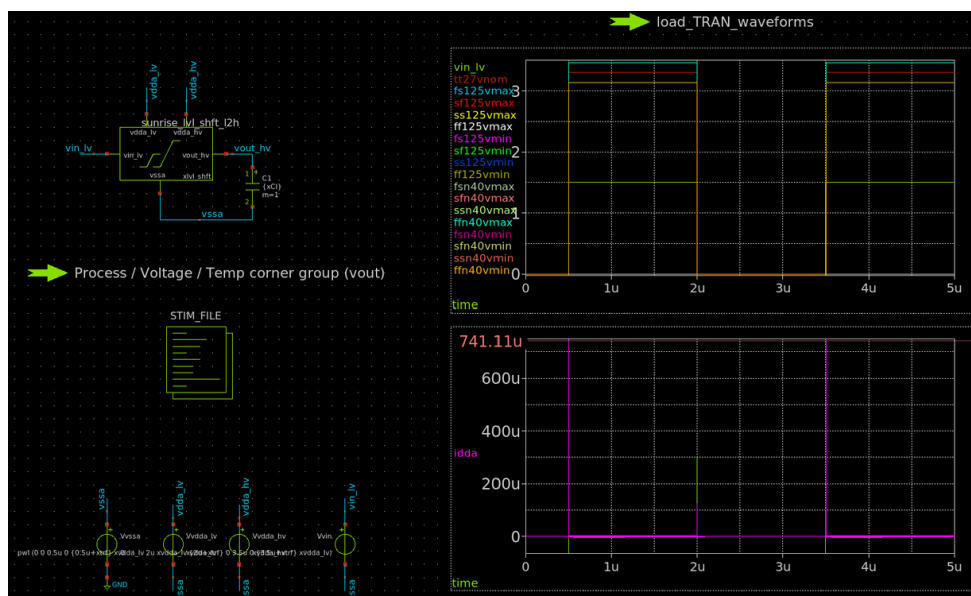


Figure 10 Testbench used to verify the level shifter core

The above git location also contains the testbench used to verify the level shifter bank and accompanying logic, whose schematic is shown in Figure 11.

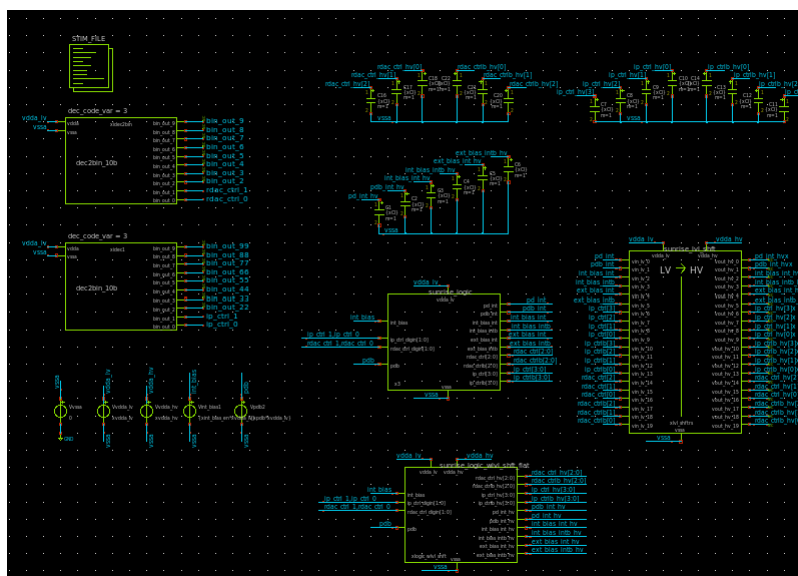


Figure 11 Testbench used to verify the level shifter bank and accompanying logic

For instructions on how to run these testbenches, the reader is referred to section 3.1 of [1].

4 Testbench simulation results (sub-blocks)

4.1 Level shifter core

4.1.1 Layout review

Layout view of the level shifter core (see Figure 4) is shown in Figure 12, where it can be seen to occupy a footprint of $\sim 5 \times 15 \mu\text{m}$.

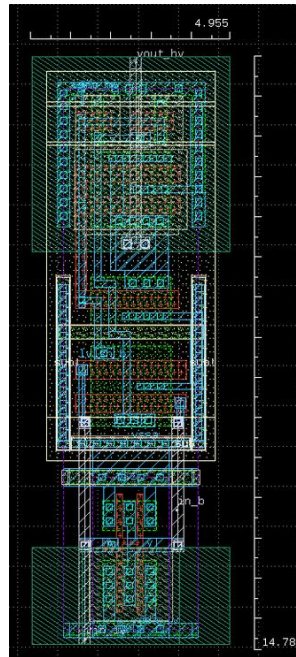


Figure 12 Level shifter core layout

One should note that various modifications were made to the layout from the original one donated from [15].

4.1.2 Functionality

Level shifter was tested by examining its output for an input stimulus which toggled low $\rightarrow$ high $\rightarrow$ low $\rightarrow$ high. As shown in Figure 13, output toggling is verified across PVT for the schematic view.

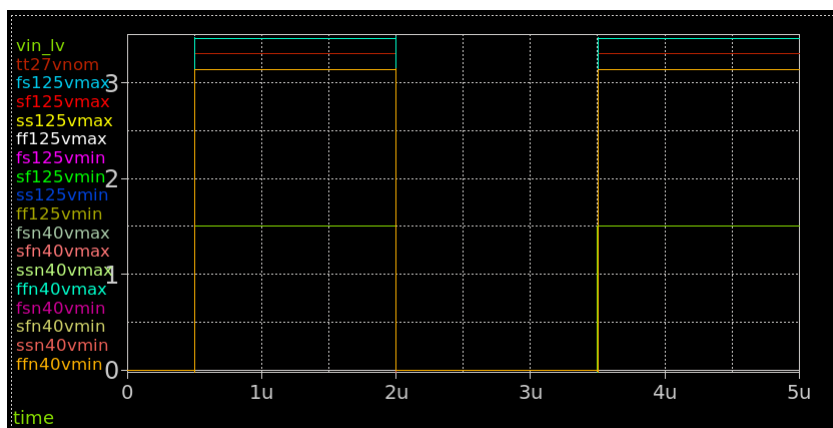


Figure 13 Level shifter schematic functionality verification across PVT

Post layout analysis was then verified by passing the same input stimulus through a full RC extracted netlist, and comparing with the schematics results at the same TT/27C corner. As shown in Figure 14, pre and post layout results match perfectly, verifying the layout to pass functionality check.

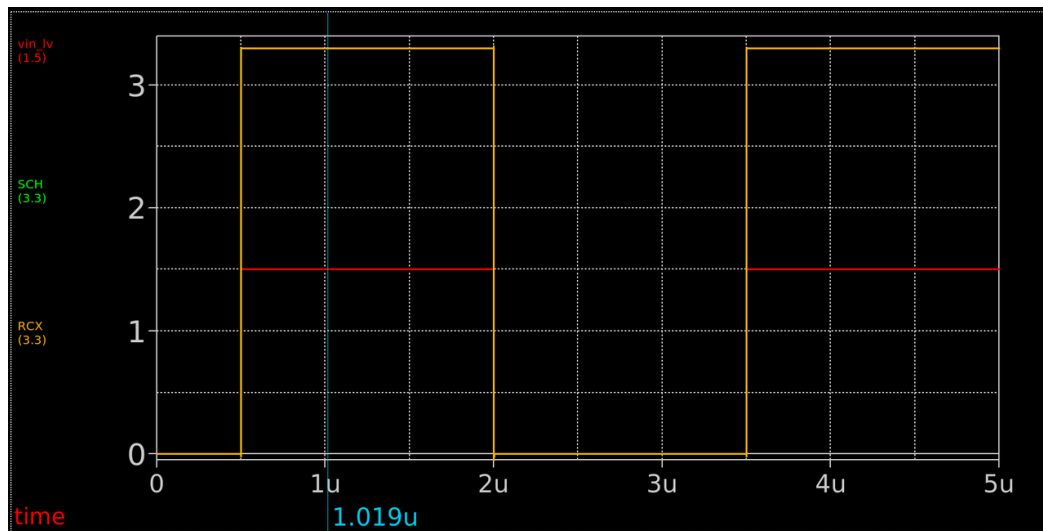


Figure 14 Level shifter pre (SCH) and post (RCX) layout functionality verification

4.2 Level shifter bank and accompanying logic

4.2.1 Layout review

The level shifter bank layout is shown in Figure 15 where it can be seen to occupy a footprint of $\sim 70 \times 40 \mu\text{m}$.

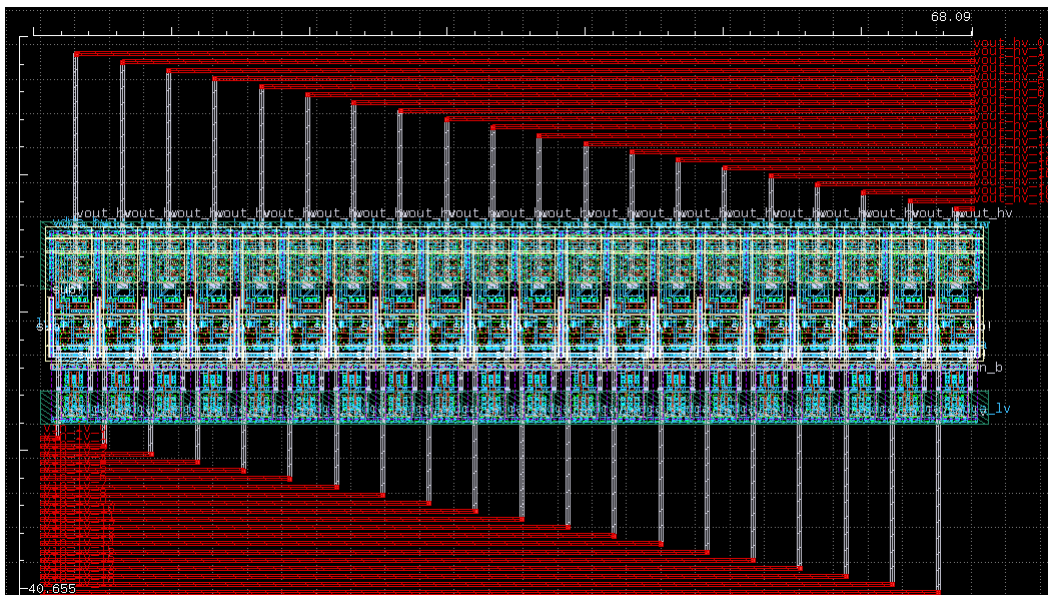


Figure 15 Level shifter bank layout

The bank with its accompanying logic is then shown in Figure 16.

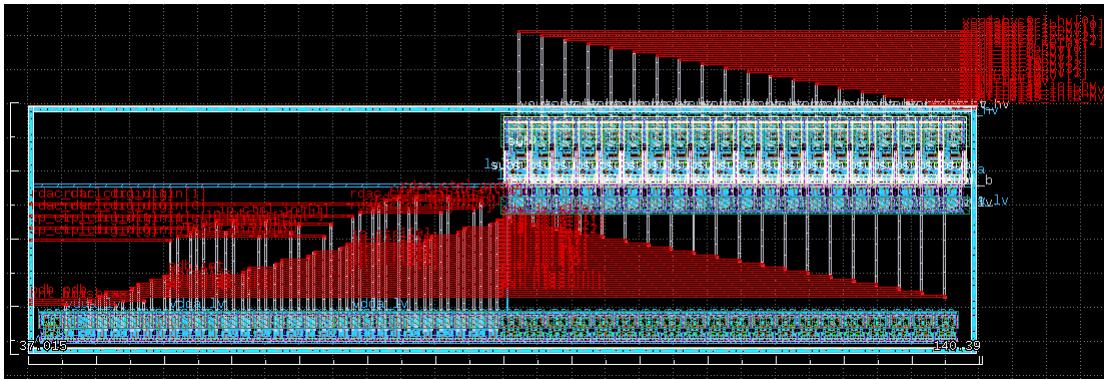


Figure 16 Level shifter bank and accompanying logic

4.2.2 Functionality

Functionality of all logic outputs is verified for schematic view in Figure 17.

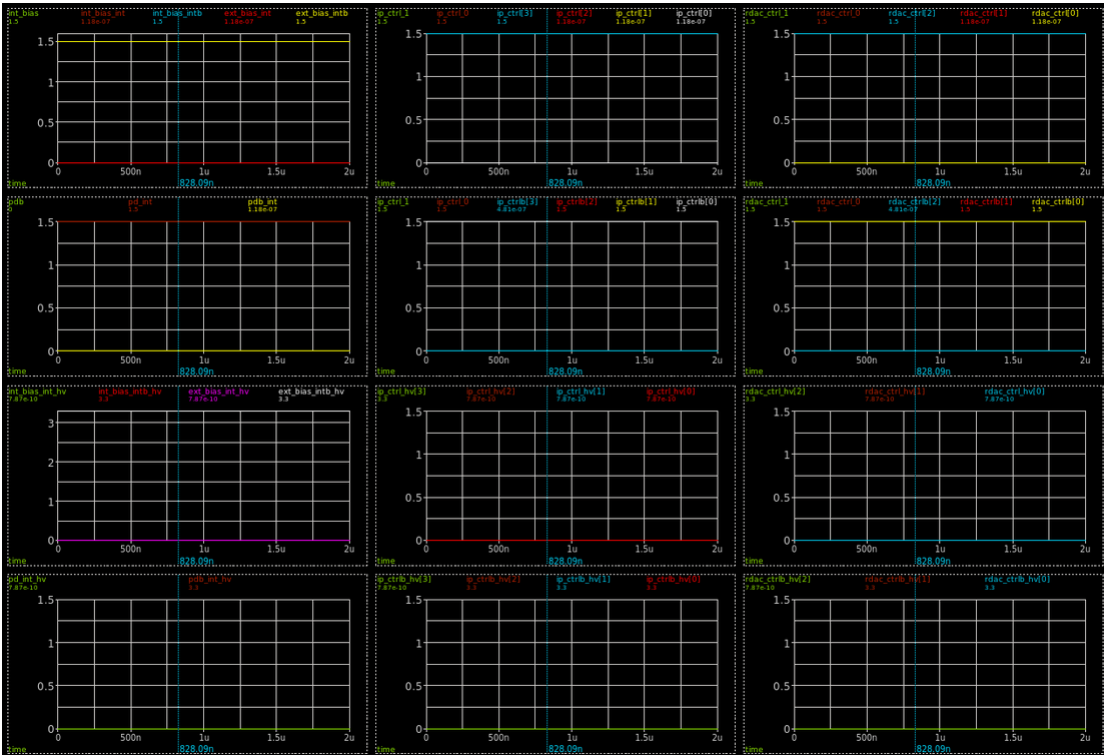


Figure 17 Level shifter and accompanying logic schematic functionality verification

Functionality of all logic outputs is verified for layout view in Figure 18. Cross inspection between the outputs in Figure 18 and Figure 17 confirm both to match perfectly, hence verifying functionality post layout.

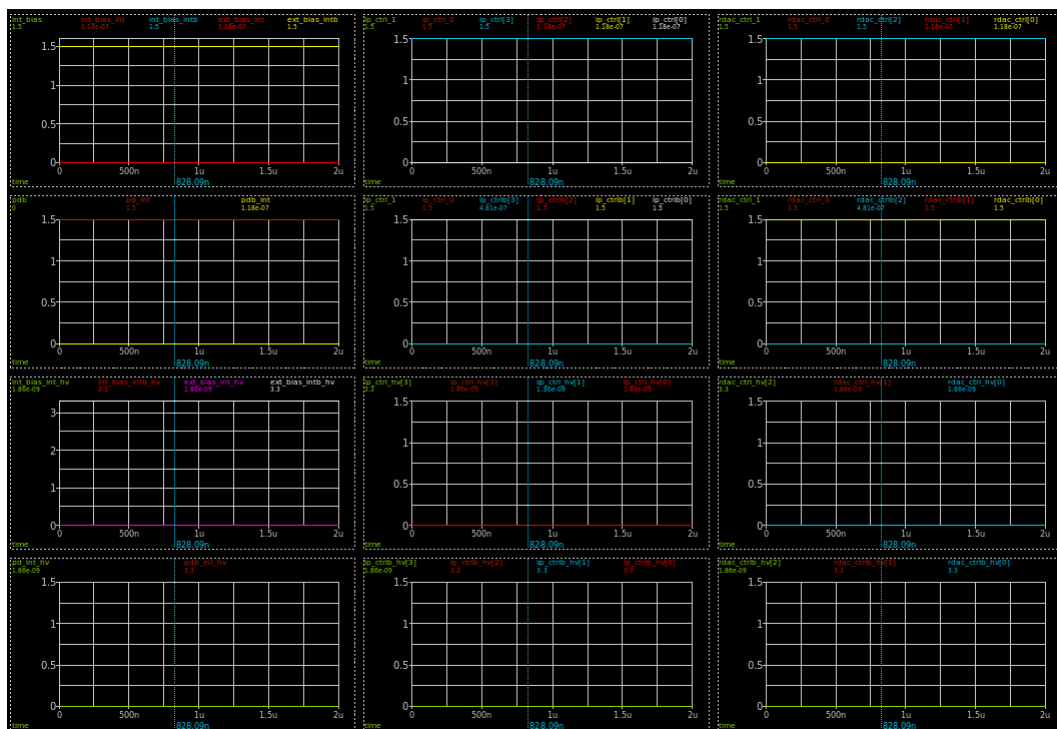


Figure 18 Level shifter and accompanying logic layout functionality verification

5 Summary

Level shifter core was imported as IP donated by Roel Jordans. Schematic was modified slightly with layout undergoing more extensive modifications. This was placed in a bank of 20 level shifters with accompanying logic that gets level shifted from vdd_{lv} to vdd_{hv} . Functionality of the level shifter core / bank and accompanying logic has been verified pre and post layout.

As the core contains no reset, it necessitates a specific power up sequence (vdd_{lv} comes up before vdd_{hv}). This has been acknowledged and reported as an ASIC requirement in [17].

6 References

- [1] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_2/INA
- [2] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_3/INA
- [3] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_2/RO
- [4] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_3/RO
- [5] <https://www.analog.com/media/en/technical-documentation/data-sheets/AD8223.pdf>
- [6] <https://ngspice.sourceforge.io/download.html>
- [7] https://xschem.sourceforge.io/stefan/xschem_man/install_xschem.html
- [8] <https://openvaf.semimod.de/>
- [9] <https://octave.org/download>

[10] <https://www.klayout.de/build.html>

[11] <https://opencircuitdesign.com/magic/download.html>

[12] <https://opencircuitdesign.com/netgen/>

[13] <https://github.com/IHP-GmbH/ihp-sg13cmos5>

[14] <https://github.com/IHP-GmbH/IHP-Open-PDK>

[15] <https://www.ihp-microelectronics.com/services/research-and-prototyping-service/mpw-prototyping-service/schedule-price-list>

[16] https://github.com/rjordans/TO_Sep2025/tree/main/FLDO/design_data

[17] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_4

[8] <https://www.ihp-microelectronics.com/services/research-and-prototyping-service/mpw-prototyping-service/schedule-price-list>

[9] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_1

[10] R. Hogervorst, J. P. Tero, R. G. H. Eschauzier and J. H. Huijsing, "A compact power-efficient 3 V CMOS rail-to-rail input/output operational amplifier for VLSI cell libraries," *Proceedings of IEEE International Solid-State Circuits Conference - ISSCC '94*, San Francisco, CA, USA, 1994, pp. 244-245

[11] <https://www.tek.com/en/products/oscilloscopes/oscilloscope-probes/passive-probe>

7 Revision History

Rev.	Date	Author	Changes, Remarks
A	11.08.2026	Diarmuid Collins	Created

Table 2: Revision history